

AI Technical Program Manager Handbook

Strategic Guide for Delivering Enterprise AI Solutions

1. AI Foundations

AI systems are probabilistic. Unlike traditional software development, AI requires managing uncertainty, data lineage, and model drift.

TPM Pro-Tip: Don't start with the model. Define the business KPI first. Does AI solve the problem better than a simple heuristic?

2-6. Generative AI, RAG, and Agents

Modern AI engineering is about orchestration. The RAG (Retrieval Augmented Generation) pattern is the standard for enterprise knowledge management.

7. AI Product Lifecycle

- Discovery & Problem Definition
- Data Governance & Ingestion
- Model Training / Fine-tuning
- Evaluation & Benchmarking
- CI/CD for AI (LLMOps)

8-16. Azure AI Stack

Component	Purpose
Azure AI Search	Semantic retrieval for RAG
Azure Functions	Orchestrating agentic workflows
Azure AI Foundry	Model evaluation and deployment

17-20. Program Management & Risk

AI projects fail due to poor data quality, scope creep, and lack of evaluation frameworks. Mitigation involves robust CI/CD, guardrails, and cost management.

21-25. Career & Leadership

As a TPM, you bridge the gap between Data Scientists and Business Executives. Proficiency in Azure, LLMOps, and Risk Management is your toolkit.